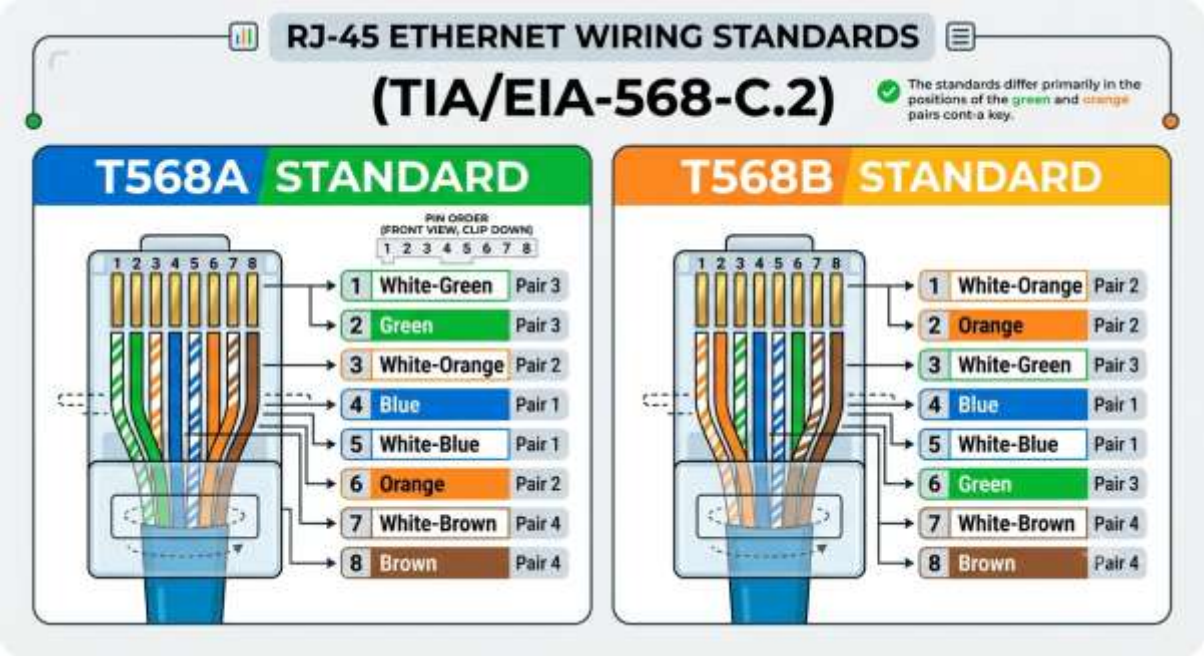
1.

Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

ANS:-



2.

Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

ANS:-

Comparison of Ethernet Standards

Specification	**10BASE-T**	**100BASE-TX**	**1000BASE-T**
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable type	Cat3 or better UTP	Cat5 UTP/STP	Cat5e or better UTP
Maximum cable length	100 metres	100 metres	100 metres
Connector type	RJ-45 (8P8C)	RJ-45 (8P8C)	RJ-45 (8P8C)
Common use	Older LANs	Basic office networks	Modern home and college networks

Note: All three standards use twisted-pair copper Ethernet cabling. 1000BASE-T uses all four wire pairs to transmit and receive data simultaneously.

3.

Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why. Hint: Think about how transmit and receive pairs are wired in each cable type.

ANS:-

Connecting Two Laptops Directly Using Ethernet

A crossover cable is traditionally used to connect two laptops directly because it swaps the transmit and receive wire pairs.

- Straight-through cable: The wire arrangement is the same at both ends. It is commonly used to connect different types of devices, such as a laptop to a switch.
- Crossover cable: The transmit and receive pairs are crossed so that one laptop's transmit pair connects to the other laptop's receive pair, and vice versa.

Conclusion: A crossover cable is the traditional choice for connecting two laptops directly. However, most modern laptops with Ethernet ports support Auto-MDI/MDI-X, which automatically detects the wiring, so a straight-through cable usually works as well.

4.

List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

ANS:-

Three Real-World Devices That Use Ethernet

Device	**Likely Ethernet standard**	**Reason**
Wi-Fi router	1000BASE-T	Provides high-speed wired internet and LAN connectivity.
Desktop computer in a college lab	1000BASE-T	Supports fast file transfers and network access.
Network printer in a college office	100BASE-TX or 1000BASE-T	Connects to the local network for shared printing.

Note: The exact standard depends on the device's hardware specifications and the capabilities of the connected network port.